

Variations

Worksheet 1 (Non-Calculator)

1. S18/qp12/q4

y is inversely proportional to the square of x . Given that $y = 10$ when $x = 3$,

find y when $x = \frac{1}{2}$. [2]

2. W18/qp11/q3

y is directly proportional to the square of x . Given that $y = 8$ when $x = 4$,

find y when $x = 3$. [2]

3. W18/qp12/q3

y is directly proportional to the square of x . Given that $y = 8$ when $x = 4$,

find y when $x = 3$. [2]

4. S19/qp12/q19

y is inversely proportional to the square of x . When $x = 4$, $y = 10$.

a. Find the value of y when $x = 10$. [2]

b. Describe the effect on y when x is halved. [1]

5. W19/qp11/q7

y is directly proportional to x . When $x = 4$, $y = t$. Find x , in terms of t , when $y = 2$. [2]

6. W19/qp12/q10

y is inversely proportional to x . When $x = 2$, $y = t$.

Find an expression for y , in terms of t , when $x = 3$. [2]

7. S20/qp11/q20

x	4	9	d
y	3	c	0.6

y is inversely proportional to the square root of x . Find the value of c and the value of d . [3]

8. W20/qp11/q5

y is directly proportional to the cube of x . When $x = 2$, $y = 4$. Find y when $x = 3$. [2]

9. S21/qp11/q19

y is proportional to $(x - 1)^2$. Given that $y = 18$ when $x = 4$, find y when $x = 6$. [2]

10. S21/qp12/q18

y is inversely proportional to the cube of x . When $x = \frac{1}{2}$, $y = 24$.

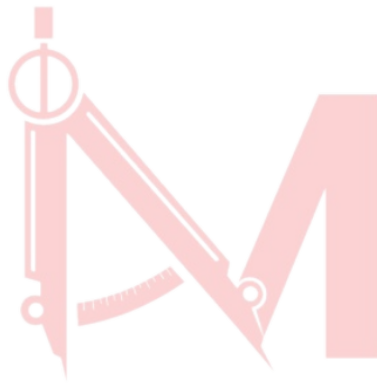
a. Find the formula for y in terms of x . [2]

b. Find the value of y when $x = \frac{1}{3}$. [1]

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Answer Key:

1. 360
2. 4.5
3. 2
4. (a)1.6 (b)y is multiplied by 4
5. $\frac{8}{t}$
6. $\frac{2t}{3}$
7. c=2 d=100
8. 13.5
9. 50
10. (a) $\frac{3}{x^3}$ (b)81



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